1 The nature of technology: Computer networks

You will learn

- how digital devices can be connected to make networks
- what the internet is and what services it provides
- how the internet helps us work together in the modern world.



A computer is a powerful tool that we use for working and learning. Connecting computers together makes them more powerful and more useful. When computers are connected, we can use them to communicate with each other. We can share files and search for information on the web. We use networks to connect computers together. In this unit you will learn how networks improve the way we live our lives.



Talk about...

Do you think you will use computers in your job when you leave school? Would you like to work in computing when you leave school? Can you think of any jobs that don't use computers?



Learning outcomes: Explain that digital devices can be connected by communication links; Explain what the internet is and some internet services such as the world wide web; Describe some ways the internet helps us work together in the modern world

network network device
local area network
wide area network
internet world wide web
Wi-Fi server hub router



Class activity

Here are two statements about how young people should use the internet.

1 "The internet is dangerous. Young people should not be allowed to use the internet."

2 "Young people should be able to use the internet whenever and wherever they choose."

Discuss the statements in your class. List the arguments for and against each statement.

Can your class write your own statement about how young people should use the internet?



Digital citizen of the future

Whatever you do in your future life, computers and networks will play an important part. The world of computing changes quickly. You will need to keep your skills up to date so you can use technology at work and at home. At work you will go on training courses to learn new skills. At home you will use the internet to learn new skills. As a good digital citizen, you will help people who have fallen behind with their computing skills.



Did you know?

Today, almost every computer is connected to a network. The biggest network in the world is the internet. The internet has the potential to connect computers wherever they are.



1.1 What is a network?

In this lesson

You will learn:

- what a computer network is
- why networks are important at school, at home and in the workplace.

Spiral back

Last year, you learned about how computers help us learn, work and enjoy our free time. Now you will learn about how computers and other devices are connected and why this is important.

What are networks for?

In offices and schools, computers are usually connected to make a computer **network**.

Networks are complicated to set up. Networks are expensive to run because they need a trained person to fix problems. Most organisations think the cost is worth it because a network has many advantages.

The advantages of networks

- **Communicating:** We can send messages and emails over the network.
- **Sharing:** We use a network to share files and expensive devices like printers.
- **Saving work:** We can save files onto network storage drives. You can use your files on any computer on the network. It is easy to share files with other people.
- **Working together:** Networks help people to work together.

What makes a network?

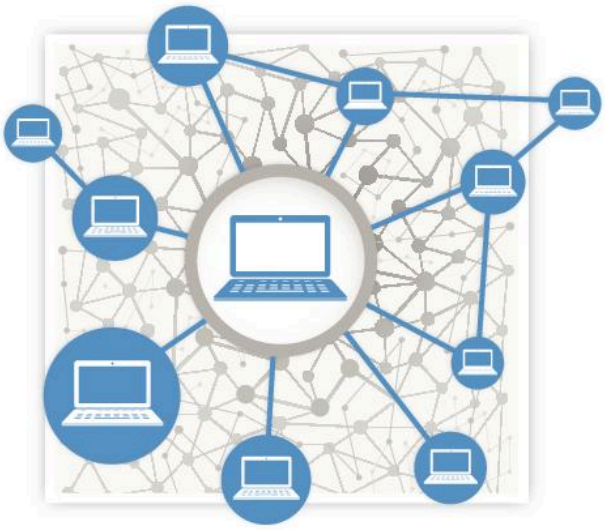
Four things are needed to make a network:

- **Network devices** to send messages and files between computers, and to store files and applications
- **Cables** to join the devices together
- **Network software** to make the devices work together
- Rules that allow all the parts of a network to work together

You will learn more about network devices and how they are connected in Lessons 1.2 and 1.3.



The two main types of network



- A **local area network** (LAN) joins computers in a single building. A LAN allows people in that building to work together. A school network is a LAN.
- A **wide area network** (WAN) joins computers that are far apart. An organisation with offices in different cities or countries uses a WAN so the employees can work together. The internet is a WAN.

The 'invisible' network

A school network can have 30 or more network devices. It can have 10 kilometres of cable running around the school.

A network is big, but most people don't notice it is there. Most of the equipment is locked away to keep it safe. But there are clues that the network exists. In this unit you will learn to spot the clues.



Activity

Web Valley Primary School has 60 network connections. The average cable used for each connection is 50 metres long. How many metres of network cable are there in the school? Convert your answer to kilometres.





Extra challenge

Use a spreadsheet to solve the problem in the activity.

Then use your spreadsheet to solve this problem:

Web Valley Secondary School has 140 network connections. The average cable is 65 metres long. How many kilometres of network cable are there in Web Valley Secondary School?

Think again

Do you save your school work to the school network? What is your personal area of the school network called? Is there an area on the network that your teacher uses to share files with the class?

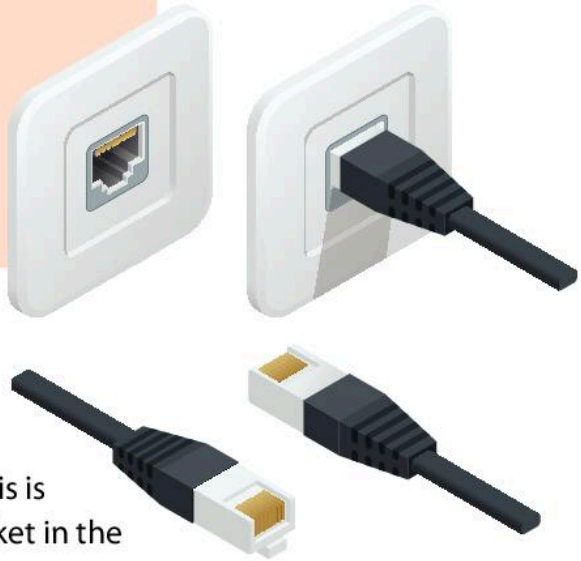
Where is the network equipment in your school stored?

1.2 Network connections

In this lesson

You will learn:

- how computers are connected to a network
- how to set a strong password and how to keep your password safe.



Connecting to a network

There are two ways to connect a computer to a network.

Wired connection

If you use a cable to plug the computer into the network, this is called a **wired connection**. A network cable connects a socket in the computer to an identical socket on the wall of a room.

If you see a socket like the one in the picture, it is a clue that the building you are in has a local area network (LAN).

Wireless (Wi-Fi) connection

You can also connect to a network using a **wireless connection**. This is also called **Wi-Fi**.

Networks use a device called a **wireless access point** (WAP) to provide a wireless connection. If you are close to a WAP, you can connect to a network without using a cable. A place where a wireless connection is available is called a hotspot.

You usually see WAPs high on the wall or on the ceiling.

Laptop and tablet computers usually use a wireless connection to a network. Desktop computers are usually connected to a network with a cable.



Using a network login

To use a network, you must have a **login**. A login is made up of a user name and a **password**.

Using a login keeps the network safe. Only people who have permission can use the network. Only you can see your own files on the network.



Keep your password safe

You must keep your password safe to protect your information and your work.

- Never tell anyone your password.
- Change your password regularly.
- Choose a strong password.

A strong password

A strong password is difficult for another person to guess. You can use the passphrase method to make a strong password.

1. Think of a short phrase that you can remember. Use two or three short words with eight to twelve characters in total. For example: **'Ginger cats'**
2. Remove the spaces between the words: **Gingercats**
3. Replace some of the letters with other characters that look similar. For example: a → @, e → 3, s → $.
4. This is your new strong password: **Ging3rc@t$**



Activity

Use the passphrase method to make a new strong password for your login.



Extra challenge

Create a poster that reminds students to keep their password safe. Include a guide for creating a strong password. Use your own example of a strong password. Can you suggest different characters to use instead of letters in the phrase?



Explore more

Logins are not just used for school networks. People use logins at work and for online shopping and banking. Ask an adult at home to help you make a list of all the things they use logins for.

Share the passphrase method of creating a strong password with the adult.